

Paper Summaries

Tyler et al. (2024)

Tyler et al. (2024) conducted a scoping review of 29 studies published between 2018 and 2023 examining the application of artificial intelligence (AI) and machine learning in emergency department triage. The review synthesised studies that applied supervised and ensemble machine learning models to predict clinical outcomes, including hospital admission, mortality risk, disease severity, and resource utilisation. Across the included studies, ensemble methods such as XGBoost were frequently reported to outperform traditional triage systems such as the Emergency Severity Index (ESI) in predictive accuracy. Despite these findings, most studies were retrospective and single-centre in design, with limited external and prospective validation, reducing confidence in real-world clinical translation.

Da'Costa et al. (2025)

Da'Costa et al. (2025) conducted a literature review of AI-driven triage systems designed to address emergency department overcrowding, delayed patient assessment, and inconsistency in triage decision-making. The review examined applications of machine learning and natural language processing models aimed at improving patient prioritisation, resource allocation, and clinical decision support. Overall, findings suggest that AI systems can enhance triage accuracy, reduce waiting times, and improve operational efficiency in emergency care settings. However, the authors highlight persistent limitations, including algorithmic bias, limited interpretability, variability in data quality, ethical concerns, and a lack of prospective evaluation in real-world clinical environments.

Araouchi and Adda (2024)

Araouchi and Adda (2024) developed TriageIntelli, a multimodal AI-assisted triage system for healthcare settings, evaluated using multiple machine learning approaches, including Support Vector Machines, Random Forests, Neural Networks, Gradient Boosting Machines, XGBoost, and a stacking ensemble method. The models were trained and tested on a single structured dataset, with performance evaluated using classification accuracy, where the stacking ensemble achieved the highest score of 80.05%. Although the results indicate strong predictive performance under controlled conditions, the system was not validated in a live clinical environment or across multiple institutions, limiting its external validity and practical deployment potential.

Literature and Identified Gaps

Gap 1: Lack of prospective evaluation of AI-assisted triage in real-world clinical workflows

Current studies demonstrate that AI-assisted triage models achieve high predictive performance in retrospective datasets or controlled experimental settings (Tyler et al., 2024; Da'Costa et al., 2025). However, these studies do not assess how AI decision-support affects workflow efficiency, triage consistency, or assessment time in live emergency department (ED) operations. Prospective evaluation in real-world settings is crucial because ED clinicians operate under time pressure, incomplete patient information, and variable patient loads, conditions that retrospective datasets cannot replicate. This gap is feasible to address in a 12-week pilot by implementing a short-term observational study in a single ED, measuring triage accuracy, inter-rater agreement, and time per patient while maintaining clinician oversight of final decisions.

Gap 2: Limited external validation and generalisability of AI triage models

Many AI triage systems are developed using single-centre or narrowly defined datasets, limiting confidence in their performance across diverse patient populations and healthcare settings (Araouchi & Adda, 2024; Tyler et al., 2024). Differences in patient demographics, triage protocols, and institutional workflows may significantly affect model accuracy and reliability. This gap can be addressed within a 12-week pilot by evaluating the AI tool on the local ED patient population and comparing model predictions against existing triage outcomes. While full multi-site validation is beyond the pilot scope, this study can generate preliminary evidence on the model's adaptability to new populations, inform future scaling studies, and highlight potential sources of variability in performance.

Draft Problem Statement

Emergency departments face rising patient demand, prolonged waiting times, and variability in triage decision-making. Although AI-assisted triage systems have demonstrated improved predictive performance compared with traditional methods, most evidence comes from retrospective studies with limited real-world evaluation. There remains insufficient understanding of how AI decision-support affects triage consistency and workflow efficiency in routine practice. This 12-week pilot will evaluate an AI-assisted triage tool to assess its impact on triage accuracy, consistency, and assessment time while maintaining clinician oversight of final decisions.

Integration of AI Tools & References

AI tools (ChatGPT) were used to support the structuring, refinement, and academic editing of this literature review. Their use included improving clarity, ensuring consistent scholarly tone, and assisting in the synthesis of recurring themes across studies, particularly limitations related to real-world validation and external generalisability. All analytical interpretations were independently reviewed, and no AI-generated content was used without manual verification or acceptance.

References

- Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D. W., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2024). *Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review*. Cureus, 16(5), e59906. <https://doi.org/10.7759/cureus.59906>
- Da'Costa, A., Teke, J., Origbo, J. E., Osonuga, A., Egbon, E., & Olawade, D. B. (2025). *AI-driven triage in emergency departments: A review of benefits, challenges, and future directions*. International Journal of Medical Informatics, 197, 105838. <https://doi.org/10.1016/j.ijmedinf.2025.105838>
- Araouchi, Z., & Adda, M. (2024). *TriageIntelli: AI-Assisted multimodal triage system for health centers*. Procedia Computer Science, 251, 430–437. <https://doi.org/10.1016/j.procs.2024.11.130>